
The kaTeX rendering for .md files is horrendus, thus here is a nicer L^AT_EX format

1 Simple Linear Regression model

We define our SLR model.

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i \quad (1)$$

Where Y_i is our response variable, β_1 and β_0 is both our parameters and ϵ_i is the random error term. We can start off with our assumptions for SLR, which are $\epsilon_i \stackrel{i.i.d}{\sim} N(0, \sigma^2)$. Since ϵ_i is a random variable with an assumed Gaussian distribution, It is obvious to see that Y_i is then a random variable from the Gaussian distribution. Such that

$$Y \sim N(\beta_0 + \beta_1 X_i, \sigma^2) \quad (1.2)$$

To find this is trivial, we take the expected value of Y_i such that

$$\begin{aligned} \mathbb{E}[Y_i] &= \mathbb{E}[\beta_0 + \beta_1 X_i + \epsilon_i] = \mathbb{E}[\beta_0] + \mathbb{E}[\beta_1 X_i] + \mathbb{E}[\epsilon_i] = \beta_0 + \beta_1 X_i \\ \mathbb{E}[Y_i] &= \beta_0 + \beta_1 X_i \end{aligned} \quad (1.3)$$

X_i is our known constant, β_0 and β_1 are the unknown constants, and ϵ_i is assumed to have $\mathbb{E}[\epsilon_i] = 0$.

1.1 Least Squares method

We are sometimes interested in the observed values deviation from its mean to distinguish the variability within the distribution of Y_i . Let Q define the least squares equation for Y_i .

$$Q = \sum_{i=1}^n (Y_i - \mathbb{E}[Y_i])^2 \quad (1.4.1)$$

A high Q indicates high variability within the observed values, this is a cause of problems for when building a model to draw predictions. We can minimize Q w.r.t the regression model parameters. In order to do this we borrow the equation from (1.3) and set the partial derivative equal to zero.

$$\frac{\partial Q}{\partial \beta_1} = -2 \sum_{i=1}^n X_i (Y_i - \beta_0 - \beta_1 X_i) = 0 \quad (1.5.1)$$

$$\frac{\partial Q}{\partial \beta_0} = -2 \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 X_i) = 0 \quad (1.5.2)$$

We apply the chain rule. From the above equations, we can derive the normal equations and the linear estimators b_1 and b_0 . We will conceal the index of the sums from here on out. Below is the first normal equation.

$$\begin{aligned} -2 \sum X_i (Y_i - \beta_0 - \beta_1 X_i) &= 0 \\ \sum (Y_i X_i - b_0 X_i - b_1 X_i^2) &= 0 \\ \sum Y_i X_i - b_0 \sum X_i - b_1 \sum X_i^2 &= 0 \\ \sum Y_i X_i &= b_0 \sum X_i + b_1 \sum X_i^2 \end{aligned} \quad (1.6.1)$$

$$\begin{aligned}
-2 \sum (Y_i - \beta_0 - \beta_1 X_i) &= 0 \\
\sum (Y_i - b_0 - b_1 X_i) &= 0 \\
\sum Y_i - nb_0 - b_1 \sum X_i &= 0
\end{aligned}$$

$$\sum Y_i = nb_0 + b_1 \sum X_i \quad (1.6.2)$$

(1.6.1) (1.6.2) are the derived normal equations.

2 Least Square Estimators

Directly from our normal equation we can derive our linear estimators, which by the Gauss-Markov Theorem, OLS estimators are BLUE - best linear unbiased estimators, among all linear unbiased estimators.

2.1 Deriving our linear estimators

$$\begin{aligned}
\sum Y_i &= nb_0 + b_1 \sum X_i \quad (\text{rearrange the equation.}) \\
b_0 &= \frac{1}{n} (\sum Y_i - b_1 \sum X_i) \\
b_0 &= \bar{Y} - b_1 \bar{X}
\end{aligned}$$

We can use this equation for b_0 to find b_1

$$\begin{aligned}
\sum Y_i X_i &= b_0 \sum X_i + b_1 \sum X_i^2 \quad (\text{replace } b_0 \text{ with its equation.}) \\
\sum Y_i X_i &= \left(\frac{1}{n} \sum Y_i - b_1 \frac{1}{n} \sum X_i \right) \sum X_i + b_1 \sum X_i^2 \\
\sum Y_i X_i &= \frac{1}{n} \sum Y_i X_i - b_1 \frac{1}{n} \sum X_i^2 + b_1 \sum X_i^2 \\
\sum Y_i X_i - \frac{1}{n} \sum Y_i X_i &= -b_1 \left(\frac{1}{n} \sum X_i^2 - \sum X_i^2 \right) \quad (\text{rearrange the equation.}) \\
b_1 &= \frac{\sum Y_i X_i - \frac{1}{n} \sum Y_i X_i}{\frac{1}{n} \sum X_i^2 - \sum X_i^2}
\end{aligned}$$